

Citation Reference Library

Ardia Health Labs — EB-1A / EB-2 NIW Immigration Evidence

All citations sourced from Ardia Health Labs white paper and pitch deck documents.

⚠ **Note:** Verify DOIs for [1] and [3] against original sources before journal submission.

NLM (National Library of Medicine) Format

[1] Georgetown University / JAMA Network Open. Next Generation Sequencing Denial Rates in Medicare. JAMA Netw Open. 2025 Apr; [vol(iss)]:e[article-id]. doi:[VERIFY]. n=29,919 claims. Key findings: 23.3% overall NGS denial rate; 27.4% post-NCD; 2.76× higher denial odds for independent vs. hospital-based laboratories.

[2] XiFin, Inc. 2024 Payor Denial Impact Report. San Diego, CA: XiFin; 2024. Analysis of 20,000,000+ laboratory claims. Key findings: Molecular diagnostics denial rate 35.3% — highest across all healthcare specialties.

[3] Franchetti MK, et al. Pharmacogenomics reimbursement outcomes in US clinical practice. Front Pharmacol. 2023;14:[article-id]. doi:[VERIFY]. Key findings: 46% of PGx claims reimbursed.

[4] American Clinical Laboratory Association (ACLA). ACLA 2024 Member Survey: Impact of PAMA Reimbursement Reductions on Independent Laboratories. Washington, DC: ACLA; 2024. Key findings: 50%+ of independent laboratory directors considering consolidation or sale if PAMA cuts take effect.

[5] Healthcare Financial Management Association (HFMA); LigoLab. Laboratory Revenue Recovery Report 2024. Westchester, IL: HFMA; 2024. Key findings: 65% of denied laboratory claims never appealed; appeal win rates 50–80.7%.

[6] Waystar Health. Form S-1 Registration Statement. Filed with SEC [2024]. AI-driven RCM market valuation \$3.5B.

[7] R1 RCM Inc. Acquisition Announcement. [2024]. \$8.9B transaction. Context: Validates scale of AI-driven RCM market; hospital-focused systems do not address independent laboratory molecular diagnostics billing.

[8] Centers for Medicare & Medicaid Services (CMS). Protected Access to Medication Act (PAMA) — Qualified Infectious Disease Product (QIDP) and Clinical Diagnostic Laboratory Test (CDLT) Reimbursement Schedule. Washington, DC: CMS; [2024 update]. Up to 15%/year reductions beginning 2027; up to 45% cumulative reduction by 2029.

[9] Texas Legislature. Senate Bill 1188: An Act Relating to Accountability for Artificial Intelligence Systems in Health Care. 89th Leg., Regular Session (2025). Effective September 1, 2025.

[10] Texas Legislature. House Bill 149: Texas Responsible Artificial Intelligence Governance Act (TRAIGA). 89th Leg., Regular Session (2025). Effective January 1, 2026. Texas is the third state to enact comprehensive AI legislation.

[11] Centers for Medicare & Medicaid Services; American Medical Association (AMA). MolDX Program: DEX Registration Requirements and CPT Code Framework for Molecular Diagnostics. [2024]. 75,000+ molecular diagnostic tests; under 200 CPT procedure codes; estimated 35%+ coding error rate.

APA (7th Edition) Format

[1] Georgetown University Medical Center. (2025, April). Next generation sequencing denial rates in Medicare: A cross-sectional analysis. *JAMA Network Open*, [vol](iss). [https://doi.org/\[VERIFY\]](https://doi.org/[VERIFY]) — n=29,919; 23.3% overall; 2.76× independent lab odds.

[2] XiFin, Inc. (2024). *2024 Payor denial impact report* [Industry analysis, 20M+ claims]. XiFin. — Molecular Dx: 35.3% denial rate.

[3] Franchetti, M. K., et al. (2023). Pharmacogenomics reimbursement outcomes in US clinical practice. *Frontiers in Pharmacology*, 14, [article]. [https://doi.org/\[VERIFY\]](https://doi.org/[VERIFY]) — 46% PGx claims reimbursed.

[4] American Clinical Laboratory Association. (2024). *ACLA 2024 member survey: Impact of PAMA reimbursement reductions on independent laboratories*. ACLA. — 50%+ labs considering exit.

[5] Healthcare Financial Management Association & LigoLab. (2024). *Laboratory revenue recovery report 2024*. HFMA. — 65% denials never appealed; 50–80.7% win rate.

[6] Waystar Health. (2024). *Form S-1 registration statement*. U.S. Securities and Exchange Commission.

[7] R1 RCM Inc. (2024). *R1 RCM acquisition announcement* [\$8.9B]. R1 RCM.

- [8] Centers for Medicare & Medicaid Services. (2024). *Protected Access to Medication Act (PAMA) CDLT reimbursement schedule*. CMS. — Up to 45% cumulative cut by 2029.
- [9] Texas Legislature. (2025). *Senate Bill 1188: Accountability for artificial intelligence systems in health care* (89th Leg.). — Eff. September 1, 2025.
- [10] Texas Legislature. (2025). *House Bill 149: Texas Responsible Artificial Intelligence Governance Act (TRAIGA)* (89th Leg.). — Eff. January 1, 2026.
- [11] Centers for Medicare & Medicaid Services & American Medical Association. (2024). *MolDX program DEX registration and CPT framework for molecular diagnostics*. CMS/AMA.
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Chicago (Author-Date) Format

- [1] Georgetown University Medical Center. 2025. "Next Generation Sequencing Denial Rates in Medicare." *JAMA Network Open* [vol] ([iss]). doi:[VERIFY].
- [2] XiFin, Inc. 2024. *2024 Payor Denial Impact Report*. San Diego: XiFin.
- [3] Franchetti, M. K. et al. 2023. "Pharmacogenomics Reimbursement Outcomes in US Clinical Practice." *Frontiers in Pharmacology* 14: [article]. doi:[VERIFY].
- [4] American Clinical Laboratory Association (ACLA). 2024. *ACLA 2024 Member Survey: Impact of PAMA Reimbursement Reductions on Independent Laboratories*. Washington, DC: ACLA.
- [5] Healthcare Financial Management Association and LigoLab. 2024. *Laboratory Revenue Recovery Report 2024*. Westchester, IL: HFMA.
- [6] Waystar Health. 2024. *Form S-1 Registration Statement*. Washington, DC: SEC.
- [7] R1 RCM Inc. 2024. *Acquisition Announcement*. Chicago: R1 RCM.
- [8] Centers for Medicare & Medicaid Services. 2024. *PAMA CDLT Reimbursement Schedule*. Baltimore: CMS.
- [9] Texas Legislature. 2025. *Senate Bill 1188*. Austin: 89th Texas Legislature.
- [10] Texas Legislature. 2025. *House Bill 149 — TRAIGA*. Austin: 89th Texas Legislature.

[11] Centers for Medicare & Medicaid Services and American Medical Association. 2024. *MolDX Program and CPT Framework for Molecular Diagnostics*. Baltimore/Chicago: CMS/AMA.

Key Data Points for Immigration Filings

Statistic	Source	Use In
35.3% molecular Dx denial rate — highest in healthcare	XiFin 2024 [2]	NIW P1, EB-1A C5
2.76× higher denial odds for independent labs	JAMA 2025 [1]	NIW P1, EB-1A C5
23.3% NGS denial rate (27.4% post-NCD)	JAMA 2025 [1]	NIW P1, Paper 2
46% PGx claims reimbursed	Frontiers Pharmacology 2023 [3]	NIW P1
65% of denied claims never appealed	HFMA/LigoLab 2024 [5]	NIW P1, EB-1A C5
50-80.7% appeal win rate	HFMA/LigoLab 2024 [5]	NIW P1
\$10-12B annual preventable revenue loss	Derived from [2][5]	NIW P1
50%+ labs considering exit if PAMA takes effect	ACLA 2024 [4]	NIW P3
Up to 45% cumulative Medicare cut by 2029	CMS PAMA [8]	NIW P1, P3
7,000+ CLIA-certified independent labs at risk	CMS 2024	NIW P3
\$3.5B Waystar IPO + \$8.9B R1 acquisition	[6][7]	NIW P1 (market validation)
TX SB 1188 — first healthcare AI law	[9]	NIW P3, Paper 3
TX TRAIGA — 3rd state AI law	[10]	NIW P3, Paper 3
75,000 molecular tests / <200 CPT codes / 35%+ error rate	CMS/AMA [11]	Paper 1, Paper 2